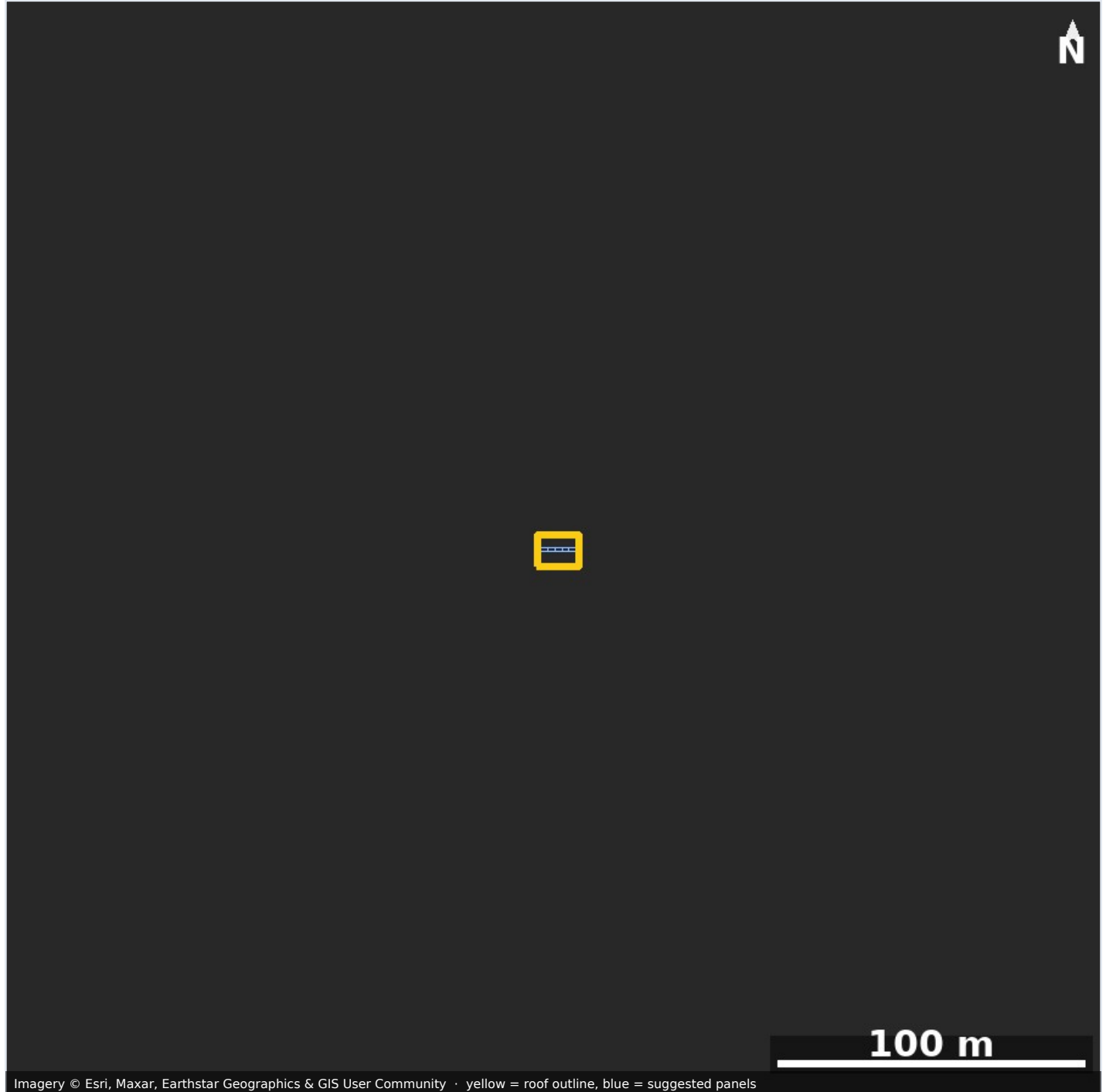




YOUR ROOF

Plot 12, Sahid Nagar, Bhubaneswar, Odisha 751007

20.29614° N, 85.82457° E · Khurda district · TPCODL



Imagery © Esri, Maxar, Earthstar Geographics & GIS User Community · yellow = roof outline, blue = suggested panels

140 m²

(1,505 sq ft roof)

Usable for solar: 98 m² (70% of roof)
Roof type: RCC / concrete flat roof (runoff 0.85)
Outline: drawn by hand on satellite map
Panel layout: 5 × 540 Wp, 0.5 m edge gap
Solar: 2.70 kW · Rainwater: 1.87 lakh L a year



GREEN SCORE

Solar 55/55 · Water 35/35 · Rule 10/10



Green Roof Report

Page 2 · Rooftop Solar report

Report #A07FE3DF

27 Sep 2026

Offline climate data



Rooftop Solar

2.70 kW · 5 panels · sized by your electricity use

3.7 units/day per kW in Khurda

SYSTEM SIZE

2.70 kW

5 × 540 Wp panels

UNITS PER YEAR

3,652

≈ 304 units / month

BILL SAVINGS / YEAR

₹17,333

≈ ₹1,444/month

YOUR BILL TODAY

₹18,470

→ ₹1,138/yr with solar

COST AFTER SUBSIDY

₹18,900

from ₹1.49 lakh before subsidy

SUBSIDY

₹1.30 lakh

₹72,600 CFA + ₹57,000 SFA

PAYBACK

1.1 years

then ~24 yrs free power

25-YEAR SAVINGS

₹4.08 lakh

net gain ₹3.89 lakh

CO₂ AVOIDED

2.6 t / year

61 t over 25 years

LIKE PLANTING

130 trees

~20 kg CO₂ per tree per year

UNITS PER KW / DAY

3.7

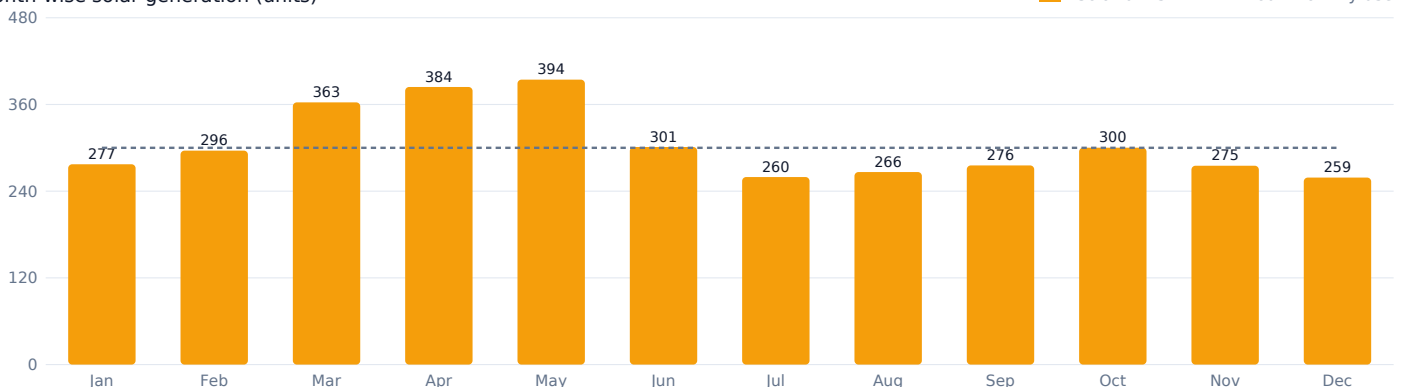
sunlight in Khurda

USABLE ROOF

98 m²

10 m² needed per kW

Month-wise solar generation (units)



How we calculated solar

Sunlight: NASA POWER (4.84 kWh/m²/day) × temperature-corrected performance ratio 0.77; 10 m² per kW (PM Surya Ghar portal); 540 Wp panels.

Money: ₹55,000/kW installed; PM Surya Ghar ₹30k/kW up to 2 kW + ₹18k for the 3rd kW (max ₹78k) plus Odisha SFA ₹25k/kW + ₹10k (max ₹60k).

Tariff: OERC domestic FY 2026-27 (₹2.90 up to 50u · ₹4.70 up to 200u · ₹5.70 up to 400u · ₹6.10 per unit, telescopic) + ₹20/kW/month fixed + 4% electricity duty; surplus settled at ₹3.59/unit (GRIDCO APPC), credited up to 90% of your yearly use.

CO₂: CEA grid factor 0.71 kg/unit; panels lose 0.5%/year; tariff held flat (conservative).

Next step: get a site survey from an MNRE/OREDA-empanelled vendor, apply on pmsuryaghar.gov.in and register for net metering with TPCODL.



Rainwater Harvesting

1573 mm rain/year in Khurda

harvest = roof area × rain × runoff 0.85

RAINWATER PER YEAR

1.87 lakh L

from 1573 mm of rain

WATER FOR

346 days

4 people at 135 L/person/day

STORAGE TANK

20,000 L

captures a 100 mm rain day

RECHARGE WELL

2 × 1 m Ø × 5.5 m

holds 8,639 L

ODISHA RULE

8,391 L

60 L/m² · met

CHHATA SUBSIDY

₹41,957

50% of ₹83,914 cost

TANKERS AVOIDED

31 / year

of 6,000 L each

DOWNPIPES

3 × 100 mm

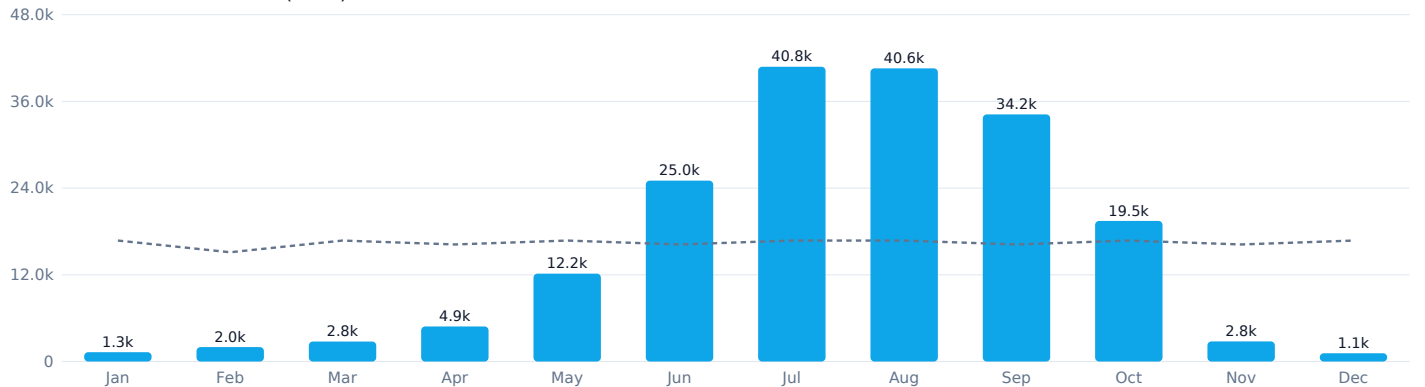
to carry the roof runoff

RUNOFF FACTOR

0.85

RCC / concrete flat roof

Month-wise rainwater harvest (litres)



How we calculated rainwater

1 mm of rain on 1 m² = 1 litre; runoff 0.85; tank sized for a 100 mm rain day; water use = 4 people (home, school or institution) × 135 L/day.

Recharge well sized to the Odisha Development Authorities Rules 2020 norm of 6 m³ per 100 m² of roof (60 L/m²).

CHHATA (Govt. of Odisha, FY 2022-23 to FY 2026-27): 50% of the system cost or ₹55,000, whichever is less, for roofs of 50–200 m² and at most 3 floors; a recharge unit is compulsory. Apply at echhata.odisha.gov.in.

Climate: Offline NASA POWER climatology for Bhubaneswar, Khurda district (0 km away), 2001–2020 average. Made with Suryajal (MobileSAM + FastAPI).



Scan to open this roof